

The Universal York Hamiltonian: An Exact Scaling Law for Constant-Mean-Curvature Time in AdS₃ Gravity, and a Stress-Tensor Dilation Law on its Boundary

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Abstract

Three-dimensional gravity with negative cosmological constant has a completely known classical solution space on a closed surface of genus $g \geq 2$, and every solution carries a canonical constant-mean-curvature (CMC) foliation whose Hamiltonian is the area of the slice. Each clock reading τ therefore defines its own Hamiltonian H_τ on the same reduced phase space $T^*\mathcal{T}_g$. We show that these are not independent. An exact, non-perturbative identity, $H_a(X, tq) = t^2 H_{at^2}(X, q)$, follows from the elliptic constraint equation defining a CMC slice and collapses the entire family onto one universal profile H_1 , sampled at a momentum scale that the clock drives to infinity within a finite range of an angular time variable. We expand H_1 at small momentum in terms of the Weil–Petersson metric and curvature, and identify its large-momentum limit with Hubbard–Masur extremal length. Reading the conjugate momentum as the classical boundary stress tensor turns the bulk law into a semiclassical stress-tensor dilation law, and yields a definite prediction for the sextic term of the boundary effective action. We state plainly which steps are theorems and which are interpretation, and we relate the proposal to recent work on volume time and TT .

1 Introduction

Once a gravitational theory is cast in Hamiltonian form, general covariance turns the Hamiltonian into a constraint, and a wavefunction annihilated by a constraint does not evolve. Physical time must therefore be manufactured from the fields themselves, by singling out some combination of them as a clock, and nothing guarantees that different clocks will agree. This is the problem of time [1, 2, 3]; it has resisted resolution largely because it is usually debated in settings too intricate to settle by direct computation.

Three-dimensional gravity with $\Lambda < 0$ is the exception. On a closed surface the classical solution space is explicitly known [5, 8, 6, 7], so the effect of a clock choice can be computed rather than argued. One slicing is singled out by geometry alone: the constant-mean-curvature (CMC) slicing of York [4], in which the clock reading τ is the trace of the extrinsic curvature of the slice, its instantaneous expansion rate. Every solution admits exactly one such foliation [9], and the Hamiltonian generating evolution in τ is, up to a constant, the area of the slice [6, 7].

A puzzle follows that we have not seen addressed directly. For each τ the construction produces a Hamiltonian H_τ on the same reduced phase space, the cotangent bundle of Teichmüller space, and there is no obvious reason for H_τ and $H_{\tau'}$ to be related by anything simpler than solving the Einstein equations twice. The naive expectation is an unbounded family of unrelated Hamiltonian systems.

The family is in fact rigid. The elliptic equation that defines a CMC slice carries an exact scaling symmetry, valid without approximation, under which H_τ is a single function H_1 of momentum evaluated at a τ -dependent scale (Section 5). The Weil–Petersson quadratic response near the time-symmetric slice and the Finsler-type extremal-length behaviour far from it, usually treated as separate features, are the two ends of that one function.

Because the momentum in question is, through the standard holographic dictionary [18, 19, 21], a classical stress tensor, the scaling law can be carried to the boundary.

Section 6 does so and states the resulting testable coefficient, but also marks where that step is interpretive rather than proved; Section 7 returns to those limits. Sections 2–4 set up the phase space, the clock formalism and the exact Hamiltonian.

2 The classical phase space of AdS₃ gravity

2.1 Mess’s parametrisation

Throughout, Σ_g is a closed, orientable surface of genus $g \geq 2$, and we set the AdS radius $\ell = 1$, so $\Lambda = -1$. Newton’s constant G and the central charge $c = 3/2G$ enter only through $16\pi G$. We write $\mathcal{T} = \mathcal{T}_g \cong \mathbb{R}^{6g-6}$ for Teichmüller space, with its Weil–Petersson (WP) Kähler metric g_{WP} [14] and symplectic form ω_{WP} [15].

Theorem 2.1 (Mess [5]; Moncrief [6]; Hosoya–Nakao [7]).
Let $\mathcal{Q} := \mathcal{T} \times \mathcal{T}$.

- (a) *Maximal globally hyperbolic AdS₃ spacetimes on $\Sigma_g \times \mathbb{R}$ are in bijection with points $(\rho_L, \rho_R) \in \mathcal{Q}$, the left- and right-moving hyperbolic structures induced at the two ends of the spacetime, with $\omega_{\mathcal{Q}} = \lambda_G(\omega_{WP}^L - \omega_{WP}^R)$ for a fixed constant λ_G .*
- (b) *Each such spacetime carries a unique foliation by constant-mean-curvature surfaces Σ_τ , labelled by $\tau = -\text{tr } K$ [9].*
- (c) *Writing $X_\tau \in \mathcal{T}$ for the conformal class induced on Σ_τ and q_τ for the holomorphic quadratic differential built from its traceless second fundamental form, the map $\Phi_\tau : (\rho_L, \rho_R) \mapsto (X_\tau, q_\tau)$ is a symplectomorphism $\mathcal{Q} \rightarrow T^*\mathcal{T}$.*
- (d) *Time reversal $\sigma(\rho_L, \rho_R) = (\rho_R, \rho_L)$ satisfies $\Phi_0 \circ \sigma = \iota \circ \Phi_0$ with $\iota(X, q) = (X, -q)$, so the time-symmetric, Fuchsian locus $\rho_L = \rho_R$ is sent to the zero section of $T^*\mathcal{T}$.*

The two holonomies ρ_L, ρ_R invite a misreading as data on one slice. They are nothing of the kind. A point of $\mathcal{Q}$ is an entire spacetime, fixed by the two hyperbolic shapes it interpolates between at its far past and far future, and Mess's theorem says that this is *all* the data there is: any pair of hyperbolic structures on Σ_g is realised by exactly one maximal globally hyperbolic spacetime (a modern account is in [11]). The phase space is thus not merely known to exist. It is an explicit, finite-dimensional object of Teichmüller theory, closely tied to the same WP geometry that controls the renormalised volume of hyperbolic 3-manifolds [10].

Part (c) trades this static description for a dynamical one, which we use from here on. The parameter $\tau = -\text{tr} K$ is the local expansion rate: $\tau > 0$ is a growing slice, $\tau < 0$ a shrinking one, and $\tau = 0$ picks out the unique momentarily static slice, the waist of the wormhole, which by (d) is exactly the Fuchsian locus. As τ runs from $-\infty$ to $+\infty$ the foliation carries us from the contracting past, through the waist, into the expanding future, and (X_τ, q_τ) records at each instant the conformal shape of the universe and the gravitational-wave shear riding on it.

2.2 The shape of a CMC slice

Fix a conformal representative ρ (the hyperbolic metric in the class X) and a holomorphic quadratic differential $q \in H^0(K_X^2)$, and define

$$\beta := \rho^{-2}|q|^2, \quad \mu := \bar{q}/\rho, \quad a := 1 + \tau^2/4.$$

The quantity β is a genuine function on Σ_g , independent of the choice of local coordinate, measuring the local density of shear. The variable $a \geq 1$ is a repackaging of the clock reading which, as Section 5 shows, is the one in which the theory becomes transparent.

Lemma 2.2 (CMC constraint equation). *A metric $h = e^{2\psi}\rho$ on Σ_g arises as the induced metric on a CMC- τ slice with shear data q if and only if ψ solves*

$$\Delta\psi = a e^{2\psi} - 1 - \frac{1}{2}\beta e^{-2\psi}, \quad (\text{E})$$

where Δ is the Laplacian of ρ .

Proof. The Hamiltonian constraint of 2+1 gravity, $R^{(2)} + (\text{tr} K)^2 - K_{ij}K^{ij} = -2$, together with the momentum constraint, which forces the traceless part B_{ij} of the second fundamental form to be (the real part of) a holomorphic quadratic differential q , reduces to (E) once the metric is written conformally as $h = e^{2\psi}\rho$; this is the reduction of [6, 7]. $\square$

Equation (E) is of Liouville type. Its first two terms are what one would write to force constant curvature in the conformal class of ρ , and the last is a source proportional to $|q|^2$. Finding a CMC- τ slice with prescribed shear is therefore the same problem as finding a constant-curvature metric in the presence of matter built from the wave content q : gravitational waves on a CMC slice curve it just as ordinary matter curves a Riemannian manifold.

Remark 2.3 (Normalisation). With β and μ defined directly from q , one has $\beta dA_\rho = |\mu|_\rho^2 dA_\rho$ exactly, so $\int_{\Sigma_g} \beta dA_\rho = \|\mu\|_2^2$ with coefficient one. No normalisation constant is introduced by hand, and every coefficient below, notably in Theorem 4.1(iii) and in the matching of Section 6, is a genuine number.

2.3 The reduced action and the York Hamiltonian

Proposition 2.4 (Reduced action; Moncrief, Hosoya–Nakao [6, 7]). *Let $p := B/16\pi G$, where B is the traceless second fundamental form built from q . Once the constraints are solved, the Einstein–Hilbert action reduces to*

$$S_{\text{red}} = \int (\langle p, \dot{X} \rangle - H_\tau) d\tau, \quad H_\tau(X, q) = \frac{A_\tau(X, q)}{16\pi G}, \quad (1)$$

where A_τ is the area of the CMC- τ slice in the metric h solving (E).

Everything below turns on this. Ordinarily a Hamiltonian attaches a number, the energy, to a configuration; here the number attached to a whole spatial slice is purely geometric, namely its own area, and the dynamics is the statement that this area changes as one slice is traded for its neighbour. No separate matter content drives the evolution: the area, as a function of the shape X and the shear q , is the entire dynamics.

3 Clocks as choices of polarisation

Before solving (E), we set up the general notion of a clock in symplectic terms. This clarifies what is special about the CMC choice, and it is this apparatus that produces H_τ as a bona fide Hamiltonian instead of a suggestively named quantity. It is a classical rendering of the observation, going back to Kuchař [2, 3], that a clock is a choice of function on phase space.

3.1 Clock atlases

Definition 3.1 (Clock atlas). Let $(\mathcal{Q}, \omega)$ be a symplectic manifold and Q a manifold. A *clock atlas* is a smooth family of symplectomorphisms $\Phi_\tau : \mathcal{Q} \rightarrow T^*Q$. Its polarisation at reading τ is the distribution $\mathcal{F}_\tau = \ker d(\pi_Q \circ \Phi_\tau)$: the directions in $\mathcal{Q}$ that the clock, at that instant, treats as leaving the reading unchanged.

The York atlas of Theorem 2.1(c), with $Q = \mathcal{T}$ and $\Phi_\tau = (X_\tau, q_\tau)$, is the running example. A polarisation decides which half of the gravitational data one is willing to call “the shape of the universe right now” and which half its instantaneous rate of change. Different clocks split the data differently, and that split, nothing more, is what generates a Hamiltonian.

Proposition 3.2 (Existence of the clock Hamiltonian). *If $H^1(T^*Q; \mathbb{R}) = 0$, as holds for $Q = \mathcal{T}$ since Teichmüller space is contractible, then for every clock atlas there is a smooth function $H_\tau \in C^\infty(T^*Q)$, unique up to the addition of a function of τ alone, such that $\partial_\tau \Phi_\tau = X_{H_\tau} \circ \Phi_\tau$.*

Proof. Set $V_\tau := (\partial_\tau \Phi_\tau) \circ \Phi_\tau^{-1}$, a time-dependent vector field on T^*Q . Each Φ_τ is symplectic, so $\mathcal{L}_{V_\tau} \omega_{\text{can}} = 0$ and $\iota_{V_\tau} \omega_{\text{can}}$ is closed; contractibility of T^*Q makes it exact, and H_τ is (minus) its primitive. $\square$

This is the familiar fact that any time-dependent canonical transformation has a generating Hamiltonian. It is pure symplectic geometry and holds for every clock atlas. The input specific to gravity is Proposition 2.4, which identifies the resulting H_τ with something computable, the slice area.

3.2 Time as non-commutativity

A clock that changes with τ carries one further piece of classical structure, which gives operational meaning to the idea that different readings disagree about “now”.

Proposition 3.3 (The clock tensor). *In Darboux coordinates (x^i, p_i) on T^*Q ,*

$$\{x_\tau^i, x_{\tau'}^j\}_Q = (\tau' - \tau) \left(\frac{\partial^2 H_\tau}{\partial p_i \partial p_j} \right)_\tau + O((\tau' - \tau)^2),$$

so that x_τ^i and $x_{\tau'}^j$ fail to Poisson-commute at first order in $\tau' - \tau$, with coefficient the Hessian of H_τ in the momenta.

Proof. Expand $x_{\tau'}^j = x_\tau^j + (\tau' - \tau) \partial H_\tau / \partial p_j + O((\tau' - \tau)^2)$ using Proposition 3.2, take the bracket with x_τ^i , and use $\{x^i, \partial H_\tau / \partial p_j\} = \partial^2 H_\tau / \partial p_i \partial p_j$. $\square$

Concretely, x_τ^i and $x_{\tau'}^j$ are the shapes two readings would each call “the universe now”, read off one space-time. That they do not Poisson-commute is the classical shadow of the fact that the two notions of “now” are tied together by the dynamics. A spacetime has one history, not a shape plus a separate rate of change, and different clocks simply slice that history differently. The momentum Hessian of H_τ measures how quickly the disagreement grows as the readings separate. It is classical and gauge-invariant, with no $\hbar$ in its definition.

4 The exact York Hamiltonian

We now establish what one needs to know about the solution of (E) without writing it down explicitly.

4.1 The variational problem

On $H^1(X)$, define

$$S_a[\psi] = \int_X \left[\frac{1}{2} |\nabla \psi|^2 + \frac{a}{2} e^{2\psi} - \psi + \frac{1}{4} \beta e^{-2\psi} \right] dA_\rho$$

$$S^*(a) := \min_{\psi} S_a[\psi].$$

Equation (E) is the Euler–Lagrange equation of S_a . The first two terms of the integrand form the ordinary Liouville functional, whose minimiser gives a constant-curvature metric in the class of ρ ; the last is the energy cost of carrying the shear. Let $G := (2 - \Delta)^{-1}$ and $\mathcal{R}(\mu) := 4\langle |\mu|^2, G|\mu|^2 \rangle_{L^2} > 0$.

Theorem 4.1 (The exact York Hamiltonian). (i)

Well-posedness. *For every (X, q, τ) , equation (E) has a unique smooth solution, the minimiser of S_a .*

(ii) **Structure.** $H_\tau = A/16\pi G$ is real-analytic in (q, τ) , satisfies $H_\tau = \frac{1}{8\pi G} \partial_a S^*$, and

$$\partial_a A = -2 \left\langle e^{2\psi}, (-\Delta + 2ae^{2\psi} + \beta e^{-2\psi})^{-1} e^{2\psi} \right\rangle < 0;$$

moreover $H_\tau(X, q) = H_\tau(X, -q) = H_{-\tau}(X, q)$.

(iii) **Small-momentum expansion.** As $q \rightarrow 0$,

$$16\pi G H_\tau = \frac{\text{Vol}}{a} + \|\mu\|_2^2 - \frac{a}{2} \mathcal{R}(\mu) + O(\|q\|^6), \quad (2)$$

where $\mathcal{R}(\mu) = -\text{Rm}_{WP}(\mu, \bar{\mu}, \mu, \bar{\mu}) = -\text{HolK}(\mu) \|\mu\|_2^4$, the holomorphic sectional curvature of the Weil–Petersson metric in the direction μ .

(iv) **Extremal-length limit.** As $|\tau| \rightarrow \infty$ with μ held fixed, $|\tau| H_\tau \rightarrow \|\mu\|_1 / 8\pi G$, proportional to the extremal length of the singular foliation associated to q by Hubbard and Masur [12].

Proof sketch. (i) The function $F(\psi) = ae^{2\psi} - 1 - \frac{1}{2}\beta e^{-2\psi}$ is strictly increasing, so a super/sub-solution and monotone-iteration argument in the style of Kazdan–Warner [13] gives a smooth solution, the maximum principle gives uniqueness, and S_a is strictly convex. (ii) Invertibility of $\Delta - F'(\psi)$ gives real-analyticity by the analytic implicit function theorem; the envelope theorem applied to $S^*(a)$ gives the first identity, and differentiating (E) in a gives the monotonicity. (iii) Expand $\psi = \psi_0 + \varepsilon_1 + \dots$ in powers of q . The linearised equation $(\Delta - 2)\varepsilon_1 = -\frac{1}{2}\beta$, together with the Tromba–Wolpert identification of the quartic term with the WP curvature tensor [16, 17], yields (2). (iv) Sandwiching $S^*(a)$ between explicit trial functions and using concavity of S^* in a gives the stated limit, recognised as a multiple of the L^1 -norm that defines extremal length [12]. $\square$

Each part has a physical reading. Part (i) says a CMC slice with given expansion and given shear always exists and is unique: there is exactly one way to embed a shape carrying a prescribed amount of wave content into the wormhole at a given instant. Part (ii) says the area falls monotonically as $|\tau|$ grows in either direction, the CMC avatar of a wormhole pinching toward past and future, and the identity $H_\tau = \frac{1}{8\pi G} \partial_a S^*$ makes the area a thermodynamic derivative of a free energy $S^*(a)$ with a in the role of an inverse temperature. Part (iii) says that near the time-symmetric slice the cost of a little shear is a positive-definite quadratic form, the WP metric itself, corrected by a quartic self-interaction set by the curvature of Teichmüller space, exactly the shape of the leading terms in a weakly self-interacting effective Hamiltonian. Part (iv) says this Riemannian picture is only a near-symmetric approximation. Far from time-symmetry, the natural size of the shear degenerates from a Euclidean-looking norm into a degree-one, Finsler-type norm tied to the singular foliation that q defines. That is as sharp a change of geometry as the passage from weak to strong field in ordinary general relativity, but it occurs here entirely inside the reduced mechanics of a fixed model.

4.2 An illustrative check

The coefficients in (2) depend only on $-\Delta + 2$ evaluated at the background solution ψ_0 and on the elementary fact that $Gf = \frac{1}{2} \int f$ for constant f . As an algebraic sanity check we minimised S_a numerically on a flat 2π -periodic torus, using 64^2 Fourier modes and a shear profile $\beta = \epsilon^2 f^2$ for a fixed test function f . The predicted quartic coefficient is reproduced to within half a percent at $\epsilon = 0.05$, with the discrepancy scaling as ϵ^6 as it should, and the ratio $A\sqrt{2a}/\int\sqrt{\beta}$, which must tend to 1 in the limit of part (iv), does so as a is taken up to 10^5 . Because the background is flat rather than hyperbolic, this tests the algebra and not the geometry; it guards against a stray sign or factor and nothing more.

5 The universal profile: an exact Euler equation for time

The technical core of the paper is that Theorem 4.1 conceals an exact scaling symmetry which collapses the family $\{H_\tau\}$ onto a single function.

Proposition 5.1 (Dilation law). *For all $t > 0$,*

$$H_a(X, tq) = t^2 H_{at^2}(X, q).$$

Proof. Substituting $\psi = \psi' + \ln t$ into S_a and rescaling $\beta \rightarrow t^2 \beta$ gives $S_a^*(t^2 \beta) = S_{at^2}^*(\beta) - \text{Vol} \cdot \ln t$; differentiating in a and using $H_a = \frac{1}{8\pi G} \partial_a S_a^*$ gives the identity. $\square$

Rescaling the shear by t can thus be compensated exactly by sending the clock parameter $a \mapsto at^2$. The mechanism is that a multiplies $e^{2\psi}$ while β multiplies $e^{-2\psi}$, so a constant shift of ψ trades one for the other. The result resembles a renormalisation-group flow with $\sqrt{a}$ as the scale.

Corollary 5.2 (The universal profile). *There is a single function $H_1(X, \cdot)$ on $T_X^* \mathcal{T}$ such that*

$$H_a(X, q) = \frac{1}{a} H_1(X, \sqrt{a} q) \quad (a \geq 1). \quad (3)$$

Proof. Set $t = \sqrt{a'}$ and evaluate Proposition 5.1 at $a = 1$. $\square$

Remark 5.3 (What H_1 is). Since $a = 1$ precisely when $\tau = 0$, the function H_1 is the Hamiltonian on the time-symmetric slice. The corollary therefore says that every clock is generated from the waist clock by a momentum rescaling. It reduces a two-parameter family to a one-parameter one; it does not say H_1 is simple, and evaluating H_1 still means solving (E) at $a = 1$.

There is no genuine family of Hamiltonians, then, but one fixed function of momentum, and all clock dependence is the kinematic question of where on its graph a given reading happens to look.

Proposition 5.4 (Euler equation for time). *H_τ satisfies the inhomogeneous Euler partial differential equation*

$$a \partial_a H_a - \frac{1}{2} q \cdot \nabla_q H_a + H_a = 0. \quad (4)$$

Proof. Differentiate Proposition 5.1 in t at $t = 1$: the left side gives $q \cdot \nabla_q H_a$, the right side gives $2H_a + 2a \partial_a H_a$; rearrange. $\square$

This is not a homogeneity statement, which would read $a \partial_a H = 0$ and would be false. It says the failure of H_τ to be independent of a is governed entirely by its radial derivative in momentum, the algebraic signature of a function that depends on its two arguments only through one combined ray variable, as Corollary 5.2 exhibits directly. The same signature appears in statistical mechanics whenever a quantity depends on two parameters only through a dimensionless ratio.

Proposition 5.5 (Trigonometric reduction of CMC time). *Let $\theta = \arctan(\tau/2) \in (-\frac{\pi}{2}, \frac{\pi}{2})$, so that $\tau = 2 \tan \theta$ and $a = \sec^2 \theta$. Then*

$$H_\tau(X, q) = \cos^2 \theta H_1(X, \sec \theta \cdot q). \quad (5)$$

Proof. Substitute $a = \sec^2 \theta$ into (3) and use $1/a = \cos^2 \theta$, $\sqrt{a} = \sec \theta$. $\square$

Writing $\Xi(\theta) := \sec \theta \cdot q$, a CMC history through (X, q) reads the fixed profile H_1 at $\Xi(\theta)$, rescaled by $\cos^2 \theta$, and for every $q \neq 0$ one has $\|\Xi(\theta)\| \rightarrow \infty$ as $\theta \rightarrow \pm \frac{\pi}{2}$, that is, as $\tau \rightarrow \pm \infty$. The clock reaches the end of time by sending the argument of a fixed function to infinity in a finite angular range, not by the function changing character. The substitution is reminiscent of the compactification used to draw a Penrose diagram, in that a divergent parameter becomes finite, but we do not identify θ with the global-time coordinate of AdS. In these cosmological spacetimes $\tau \rightarrow \pm \infty$ is the approach to the initial and final singularities, not to a timelike conformal boundary, a distinction that matters in Section 6.

Corollary 5.6 (One function, two regimes). *Substituting $q = \Xi/\sqrt{a}$ into (2) and using the quartic homogeneity of $\mathcal{R}$,*

$$16\pi G H_1(X, \Xi) = \text{Vol} + \|\mu_\Xi\|_2^2 - \frac{1}{2} \mathcal{R}(\mu_\Xi) + O(\|\Xi\|^6) \quad (\Xi \rightarrow 0), \quad (6)$$

independent of a , as it must be; and substituting the extremal-length asymptotics of Theorem 4.1(iv) into (3) with $a \approx \tau^2/4$ gives

$$16\pi G H_1(X, \Xi) \rightarrow \|\Xi\|_1 \quad (\|\Xi\| \rightarrow \infty). \quad (7)$$

Proof. For (6), substitute $q \rightarrow \Xi/\sqrt{a}$, $\mu \rightarrow \mu_\Xi/\sqrt{a}$ into (2), use $\mathcal{R}(\mu_\Xi/\sqrt{a}) = \mathcal{R}(\mu_\Xi)/a^2$, multiply through by a , and use $H_1(X, \Xi) = a H_a(X, \Xi/\sqrt{a})$: every explicit power of a cancels. For (7), substitute $|\tau| \approx 2\sqrt{a}$ and $\|\mu_\Xi\|_1 = \sqrt{a} \|\mu\|_1$ into the limit of Theorem 4.1(iv). $\square$

The WP regime and the Teichmüller–Finsler regime are therefore the near-origin and far-field behaviour of one function on the vector space $T_X^* \mathcal{T}$, and the crossover marks the scale at which the two asymptotics cease to agree. Ordinary relativity offers a fair analogy: it reduces to Newtonian gravity in weak fields and shows horizons in strong ones, yet remains a single theory. What the corollary does *not* supply is any control of the intermediate region, where neither asymptotic form is available.

6 The boundary reading: a stress-tensor dilation law

We now carry the exact bulk result to the boundary. No new bulk computation enters. What changes is that the scaling law, read through the holographic dictionary, becomes a statement about how the dual CFT₂'s generating functional must depend jointly on the stress tensor and the clock reading.

6.1 The holographic dictionary for CMC data

The momentum $p = B/16\pi G$ conjugate to X in (1) is identified in the standard dictionary with the holomorphic component of the Brown–York stress tensor [18], made finite for asymptotically AdS spacetimes by holographic renormalisation [19, 20]. In the Teichmüller language of

this paper, Scarinci and Krasnov [21] show that the holographic charges of an AdS_3 spacetime, mass and angular momentum among them, are periods of the quadratic differential q of Theorem 2.1(c). Since $B_{zz} = q$ by Lemma 2.2, q is, up to the normalisation supplied by holographic renormalisation, the classical stress tensor T of the dual CFT_2 in the state and background complex structure described by the CMC slice.

One refinement is needed. Under conformal maps $T(z)$ shifts by a Schwarzian derivative, so strictly it is a projective connection, and only its difference from a reference projective connection, for instance the Fuchsian one, is a holomorphic quadratic differential. We therefore read q as the stress tensor measured relative to that reference. Modulo the Schwarzian shift, such objects form a coadjoint orbit of the Virasoro group [22, 23], and the CMC construction then supplies an explicit τ -dependent family of points on such an orbit together with an exact formula for how the associated gravitational free energy varies along it.

6.2 The Stress-Tensor Dilation Theorem

Granting the dictionary, Corollary 5.2 translates at once into a boundary statement.

Theorem 6.1 (Stress-tensor dilation). *Let $\mathcal{E}(X, T) := 16\pi G H_1(X, T)$, with T identified with q under the dictionary above. Then, exactly on the bulk side,*

$$16\pi G H_\tau(X, T) = \frac{1}{1 + \tau^2/4} \mathcal{E}\left(X, \sqrt{1 + \tau^2/4} T\right). \quad (8)$$

At the semiclassical level (large central charge $c = 3/2G \rightarrow \infty$, where the on-shell bulk action computes the leading saddle of the CFT generating functional in a heavy state of stress tensor T), the family of CMC-time free energies collapses onto a single boundary functional $\mathcal{E}(X, \cdot)$, sampled at a stress-tensor scale $\sqrt{1 + \tau^2/4} T$ that grows with the clock reading.

We should be exact about what is proved. Equation (8) is Corollary 5.2, relabelled, and contributes no new mathematics. The added claim is physical: that $\mathcal{E}$ computes a genuine semiclassical CFT quantity, the Legendre transform of the generating functional with respect to the boundary complex structure, at fixed stress tensor rather than fixed background metric. That claim inherits everything from the dictionary and is an interpretation of an exact bulk formula, not a derivation from the CFT side.

Read that way, (8) says the semiclassical free energy, as a function of the stress tensor at fixed conformal structure, is not an arbitrary function of two variables. It is one function of a single combined variable, and the clock reading merely records which value a given saddle occupies. The dilation acts on stress-tensor phase space at fixed surface. It is not the CFT's spacetime dilatation operator, which acts on coordinates and local operators, and the two should not be conflated; the object here is best called a Virasoro-coadjoint dilation.

6.3 One boundary function, two regimes

Corollary 5.6, read through the same dictionary, becomes a statement about $\mathcal{E}$:

$$\mathcal{E}(X, T) = \text{Vol}(X) + \|T\|_{WP}^2 - \frac{1}{2} \mathcal{R}(T) + O(\|T\|^6) \quad (\|T\| \rightarrow \infty)$$

$$\mathcal{E}(X, T) \longrightarrow \|T\|_1 \quad (\|T\| \rightarrow \infty).$$

For a state whose stress tensor is small against the natural WP scale, the free energy responds quadratically, with the WP metric acting as the normalisation of the stress tensor's two-point function, corrected at the next order by the curvature of Teichmüller space. Since the WP metric is tied to the Liouville action and to the central charge by the theorems of Belavin–Knizhnik and Zograf–Takhtajan [25, 24], the quadratic coefficient is a consistency check on the dictionary, not new evidence for it. For a heavy state the same free energy degenerates into a degree-one, Finsler-type functional governed by the extremal-length geometry of T ; the boundary theory then organises itself around the measured foliations of Thurston's compactification [12] instead of any Riemannian distance in stress-tensor space.

6.4 A sharp, falsifiable prediction

The two limits above are fixed by kinematics on both sides of the duality, whether or not $\mathcal{E}$ and the true CFT effective action agree beyond leading order. The first place the two could genuinely part company is the quartic curvature term beyond the coefficient fixed by the central charge and, more sharply, the sextic term in $\|T\|$, which no symmetry argument we know of determines on either side. On the bulk side it requires one further iterate of the Kazdan–Warner method [13] used to prove Theorem 4.1. On the boundary side it requires an independent semiclassical CFT computation, for instance from large- c Virasoro blocks or heavy-state techniques, or from the exact 3d-gravity partition functions of Virasoro TQFT [30, 31] in their semiclassical limit. Agreement would be a quantitative confirmation that the bulk scaling law and the boundary generating functional are one object; disagreement would locate the point where the correspondence between q and the quantum stress tensor first fails. The test presupposes a definition of the boundary effective action that is independent of field redefinitions, a point we return to below.

6.5 Relation to volume time and $T\bar{T}$

The proposal sits beside independent work on radial evolution. Blacker, Callebaut, Hergueta and Ning [29] show that York time has a canonically conjugate volume time and that the resulting radial canonical formalism admits a semiclassical interpretation in terms of a $T\bar{T}$ -deformed CFT [26, 27, 28]. The two lines of work approach related questions from opposite ends: theirs places York time within the $T\bar{T}$ literature, while ours supplies a closed-form Hamiltonian together with the scaling law it must obey. The profile H_1 is then a candidate for the finite-cutoff, $T\bar{T}$ -deformed free energy, and (8) would describe how that free energy depends jointly on cutoff and state. We regard this as a suggestion to be tested, not a result (see Section 7).

6.6 Scope of the claims

Three cautions apply. First, the identification of q with the boundary stress tensor is exact only in the sense that holographic renormalisation makes precise; the cited results concern holographic charges as periods of q , not scheme-independence of every component of q at every τ .

Second, everything here belongs to the classical, large- c limit and is silent about $1/c$ corrections, which come from bulk loops. Third, the dilation in (8) acts on the stress-tensor phase space at fixed surface, not as the spacetime dilatation generator. A fourth, more serious, limitation concerns the domain of the dictionary itself and is taken up in the next section.

7 Discussion

The bulk result is a clean, exact statement: a CMC clock does not supply a new Hamiltonian at each tick but rereads a single one at a growing momentum scale, and the WP and Finsler regimes are two views of that function. How much of this is deep should be weighed soberly, and the holographic extension deserves more scepticism than the bulk statement.

The scaling law is exact but shallow. The identity of Proposition 5.1 exists because τ enters (E) only through the constant a multiplying $e^{2\psi}$, so that a constant shift of ψ trades a against β . It is closer to dimensional analysis than to a hidden symmetry of the dynamics, and there is no reason to expect it to survive matter, other dimensions, or the punctured and conical cases where CMC existence is itself delicate. Moreover, H_1 is just the $\tau = 0$ Hamiltonian, so the law compresses the family without computing anything: to know H_1 at all momenta one must still solve (E), and the parts of the profile of greatest interest, the crossover between the WP and Finsler regimes, lie outside both asymptotic expansions. The claim that the two regimes are “one function” is true, but the interesting content would be a rate, and none is given.

The proofs and checks are thinner than the statements. Parts (iii) and (iv) of Theorem 4.1 are sketched, not proved in full, and the only numerical check is on a flat torus, which does not solve the same equation as a hyperbolic background. A genuine test would minimise S_a on a hyperbolic surface of genus two, where the WP curvature term can be compared with known values.

The dictionary is the weak point. Brown–York and the holographic-renormalisation results are formulated for a timelike boundary at large radius. The momentum q here, by contrast, is the shear on a spacelike Cauchy slice, and the Mess spacetimes on a closed Σ_g have no asymptotic timelike boundary: $\tau \rightarrow \pm\infty$ reaches singularities. The identification of q with a CFT stress tensor is therefore not a direct application of the standard dictionary. It requires an analytic-continuation or Euclidean reading, plausibly in the spirit of the Virasoro TQFT treatment [30, 31], in which Σ_g appears as the boundary of a 3-manifold, and until that reading is made precise the phrase “dual CFT₂” should be treated as suggestive. The Schwarzian subtlety of Section 6 is a further reminder that q and T are not literally the same object.

The prediction is only as sharp as its definitions. The sextic coefficient can be compared only once the boundary effective action is defined without ambiguity; effective actions on moduli space are not canonical and

can be reshuffled by field redefinitions. A prediction that depends on scheme is not yet falsifiable. Fixing the definition, perhaps through the genus- g partition function at large c , is a precondition for the test, not a footnote.

The link to $T\bar{T}$ is a hypothesis. Radial $T\bar{T}$ evolution uses timelike slicings, while CMC slicing is spacelike, so the passage between them is not automatic. There is also a concrete tension to be explained: the $T\bar{T}$ deformation of energies has a characteristic square-root structure with a complex branch at large energy [27, 26], whereas H_1 grows linearly in the L^1 sense at large momentum. A serious version of the correspondence would have to reconcile these.

What survives is a modest but real result. The exact reduction of the CMC Hamiltonian family, the Euler equation, and the two limiting forms are checkable on their own terms, and we would welcome their extension or refutation. The holographic reading is best held as a programme with a definite target rather than an established correspondence, and the sextic coefficient, once properly defined, is the measurement that would decide it.

References

- [1] B. S. DeWitt, *Quantum theory of gravity. I. The canonical theory*, Phys. Rev. 160 (1967) 1113.
- [2] K.uchař, *Time and interpretations of quantum gravity*, Proc. 4th Canadian Conf. on General Relativity and Relativistic Astrophysics (1992).
- [3] C. Isham, *Canonical quantum gravity and the problem of time*, arXiv:gr-qc/9210011.
- [4] J. W. York, *Role of conformal three-geometry in the dynamics of gravitation*, Phys. Rev. Lett. 28 (1972) 1082.
- [5] G. Mess, *Lorentz spacetimes of constant curvature*, Geom. Dedicata 126 (2007) 3 (preprint, 1990).
- [6] V. Moncrief, *Reduction of the Einstein equations in 2+1 dimensions to a Hamiltonian system over Teichmüller space*, J. Math. Phys. 30 (1989) 2907.
- [7] A. Hosoya, K. Nakao, *(2+1)-dimensional pure gravity for an arbitrary closed initial surface*, Class. Quantum Grav. 7 (1990) 163.
- [8] E. Martinec, *Soluble systems in quantum gravity*, Phys. Rev. D30 (1984) 1198.
- [9] T. Barbot, F. Béguin, A. Zeghib, *Constant mean curvature foliations of globally hyperbolic spacetimes locally modelled on AdS_3* , Geom. Dedicata 126 (2007) 71.
- [10] K. Krasnov, J.-M. Schlenker, *On the renormalized volume of hyperbolic 3-manifolds*, Geom. Dedicata 126 (2007) 187.
- [11] F. Bonsante, A. Seppi, *Anti-de Sitter geometry and Teichmüller theory*, in *In the Tradition of Thurston*, Springer (2020).
- [12] J. Hubbard, H. Masur, *Quadratic differentials and foliations*, Acta Math. 142 (1979) 221.
- [13] J. Kazdan, F. Warner, *Curvature functions for compact 2-manifolds*, Ann. Math. 99 (1974) 14.

- [14] L. Ahlfors, *Some remarks on Teichmüller's space of Riemann surfaces*, Ann. Math. 74 (1961) 171.
- [15] S. Wolpert, *On the Kähler form of the moduli space of once punctured tori*, Ann. Math. 117 (1983) 207.
- [16] A. Tromba, *On a natural affine connection on the space of almost complex structures and the curvature of Teichmüller space with respect to its Weil–Petersson metric*, Manuscripta Math. 56 (1986) 475.
- [17] S. Wolpert, *Chern forms and the Riemann tensor for the moduli space of curves*, Invent. Math. 85 (1986) 119.
- [18] J. D. Brown, J. W. York, *Quasilocal energy and conserved charges derived from the gravitational action*, Phys. Rev. D47 (1993) 1407.
- [19] V. Balasubramanian, P. Kraus, *A stress tensor for anti-de Sitter gravity*, Commun. Math. Phys. 208 (1999) 413, [arXiv:hep-th/9902121](#).
- [20] S. de Haro, K. Skenderis, S. N. Solodukhin, *Holographic reconstruction of spacetime and renormalization in the AdS/CFT correspondence*, Commun. Math. Phys. 217 (2001) 595, [arXiv:hep-th/0002230](#).
- [21] C. Scarinci, K. Krasnov, *The universal phase space of AdS_3 gravity*, Commun. Math. Phys. 322 (2013) 167, [arXiv:1111.6507](#).
- [22] E. Witten, *Coadjoint orbits of the Virasoro group*, Commun. Math. Phys. 114 (1988) 1.
- [23] A. Alekseev, S. Shatashvili, *Path integral quantization of the coadjoint orbits of the Virasoro group and 2D gravity*, Nucl. Phys. B323 (1989) 719.
- [24] P. Zograf, L. Takhtajan, *On uniformization of Riemann surfaces and the Weil–Petersson metric on Teichmüller and Schottky spaces*, Math. USSR Sb. 60 (1988) 297.
- [25] A. A. Belavin, V. G. Knizhnik, *Algebraic geometry and the geometry of quantum strings*, Sov. Phys. JETP 64 (1986) 214.
- [26] A. B. Zamolodchikov, *Expectation value of composite field $T\bar{T}$ in two-dimensional quantum field theory*, [arXiv:hep-th/0401146](#).
- [27] F. A. Smirnov, A. B. Zamolodchikov, *On space of integrable quantum field theories*, Nucl. Phys. B915 (2017) 363, [arXiv:1608.05499](#).
- [28] L. McGough, M. Mezei, H. Verlinde, *Moving the CFT into the bulk with $T\bar{T}$* , JHEP 04 (2018) 010, [arXiv:1611.03470](#).
- [29] M. J. Blacker, N. Callebaut, B. Hergueta, S. Ning, *Radial canonical AdS_3 gravity and $T\bar{T}$* , JHEP 01 (2025) 092, [arXiv:2406.02508](#).
- [30] S. Collier, L. Eberhardt, M. Zhang, *Solving 3d gravity with Virasoro TQFT*, SciPost Phys. 15 (2023) 151, [arXiv:2304.13650](#).
- [31] S. Collier, L. Eberhardt, M. Zhang, *3d gravity from Virasoro TQFT: holography, wormholes and knots*, SciPost Phys. 17 (2024) 134, [arXiv:2401.13900](#).